Manuel Eduardo Santiago Feria

Oaxaca, Mexico • Open to Remote / Relocation

Backend Developer | Software Engineer (Node.js / NestJS / TypeScript / PostgreSQL)

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Portfolio: portfolio.nano-service.com

PROFESSIONAL SUMMARY

Computer Systems Engineering Graduate specializing in Backend Architecture and Full-Stack Development. Proven track record of building production-ready, scalable RESTful APIs and real-time web applications using **NestJS, Node.js, TypeScript, PostgreSQL, and Redis**. Strongly focused on data integrity, concurrency control, automated testing (**1,600+ unit & integration tests across projects**), and secure containerization with Docker.

TECHNICAL SKILLS

Backend & Architecture: Node.js, NestJS, TypeScript, Go, PHP (Laravel), Python, RESTful APIs, WebSockets (Socket.io), Microservices, BullMQ, JWT / 2FA.

Databases, ORM & Storage: PostgreSQL (GIN Indexes / Full-Text Search), Redis (Cache & Queues), MySQL, TypeORM, Prisma ORM, MinIO / S3 Object Storage.

Testing, DevOps & Cloud: Jest (Unit & E2E Testing), Pest PHP, Docker (Distroless / Compose), Google Cloud Platform (GCP), Nginx, Linux, Git, CI/CD.

Frontend & AI Integration: React 19, TanStack Query, Vue 3, Tailwind CSS, Vite, YOLOv8, ONNX Runtime DirectML, OpenCV.

PROFESSIONAL EXPERIENCE

Backend & Full-Stack Developer — Professional Residency

Dec 2025 — Aug 2026

Instituto Tecnológico de Oaxaca (TecNM / ITO) • IT Helpdesk, Asset Inventory & Real-Time Support System

`NestJS • TypeScript • PostgreSQL • Redis • TypeORM • Socket.io • MinIO S3 • React • Docker • Jest`

- **Modular Backend & Real-Time Architecture:** Engineered a modular REST API in NestJS and real-time bidirectional communication via WebSockets (Socket.io) to centralize IT support tickets, chatbot workflows, and hardware asset tracking with Role-Based Access Control (RBAC).
- **Data Integrity & Concurrency Control:** Implemented atomic database transactions (TypeORM `QueryRunner`) and optimistic locking (`@VersionColumn`) to prevent race conditions and data loss when multiple technicians update tickets and inventory simultaneously.
- **High-Performance Search & Caching:** Optimized hardware and ticket lookups using PostgreSQL Full-Text Search (`tsvector` and GIN indexes), paired with Redis in-memory caching, rate limiting, and S3-compatible object storage (MinIO) for ticket attachments.
- **Software Quality & Hardened Deployment:** Guaranteed production stability by writing **998 automated unit and integration tests across 143 test suites using Jest**, deploying the application in non-root Docker Distroless containers behind an Nginx reverse proxy.

KEY ENGINEERING PROJECTS

Enterprise Core SaaS Platform — Backend & Web Architecture

2025 — 2026

High-Security Modular SaaS with Asynchronous Task Processing & Automated Testing

`NestJS • TypeScript • Prisma 7 • PostgreSQL • Redis • BullMQ • React 19 • ClamAV • Jest (627 tests)`

- **Security & Background Processing:** Designed a multi-tenant SaaS backend featuring decoupled 2FA/TOTP authentication, granular RBAC permissions, asynchronous background job queues with BullMQ/Redis, S3 file storage with automated ClamAV antivirus scanning, and a test suite of **627 automated tests across 119 suites in Jest**.

AI Smart POS — Computer Vision Point of Sale System

2024 — 2025

Automated Product Detection, Multi-Object Tracking & GPU-Accelerated Inference

`Python • YOLOv8 • ONNX Runtime DirectML • ByteTrack • OpenCV • NumPy`

- **Real-Time Inference Pipeline:** Built an automated checkout system by training a custom YOLOv8 model achieving **97.1% mAP@50** on validation datasets; exported to ONNX Runtime with DirectML GPU acceleration (~21 FPS) and integrated ByteTrack object tracking to prevent duplicate billing in real time.

Reactive CMS & Role-Based Publishing Platform

2024

Full-Stack Content Management System with Server-Side Reactivity & Automated Testing

`Laravel • PHP • Livewire Volt • MySQL • Spatie RBAC • Pest PHP (44 tests)`

- **Backend & Automated QA:** Developed a content management platform featuring server-driven reactivity (Livewire Volt), fine-grained role/permission control (Spatie RBAC), automated CSV report generation, and **44 automated feature tests using Pest PHP**.

EDUCATION

B.S. in Computer Systems Engineering

Instituto Tecnológico de Oaxaca (TecNM / ITO)

2021 — 2026 • Graduated (Residency Completed)

Languages: Spanish (Native) • English (B1/B2 — Technical Reading, Documentation & Professional Communication).

VERIFIED CERTIFICATIONS

Google Cloud Computing Foundations

Google Cloud • Credly Badge: afaf62b7 • 2026

Cloud Security and Operations

Google Cloud • Credly Badge: 88ab965e • 2026